Element F: Design Viability

Risk Mitigation Plan:

Identification and Prioritize

1. Not having enough time to complete the project - Time is a crucial factor in our project. Our design mainly focuses on combining engineering and art, which both take time.
2. Low performance - We understand that our limited knowledge on engineering tasks, tools, and actions may affect our prototyping process. It could cause us to have set backs, depending on what we struggle with.
3. High material costs - Our cost of materials can affect the outcome of the project.
4. The prototype being too heavy for magnets to hold up - Our prototype will consist of wood, paper, the lightbox, and polycarbonate glass. Depending on our final design, the weight of all the wood pieces combined would be around 2-3 lbs, the lightbox would be about 1 lbs, and the polycarbonate glass would be about 1 lbs.

Performing a Risk Assessment

All of these events can have a high likelihood of happening. Most of these events connect and result in our team not having enough time to fully complete and perfect our prototype.

Track risks

1. Monitoring a daily progress journal to ensure that everyone is doing their part in the project.
2. Being extra careful of the engineering supplies (saws, hammers, cutters) to avoid injuries, studying the related skills and techniques needed to ensure the success of the tools.
3. Taking advantage of every resource we have to reduce the amount of money spent on the project.
